Points 6 Published

Details Questions

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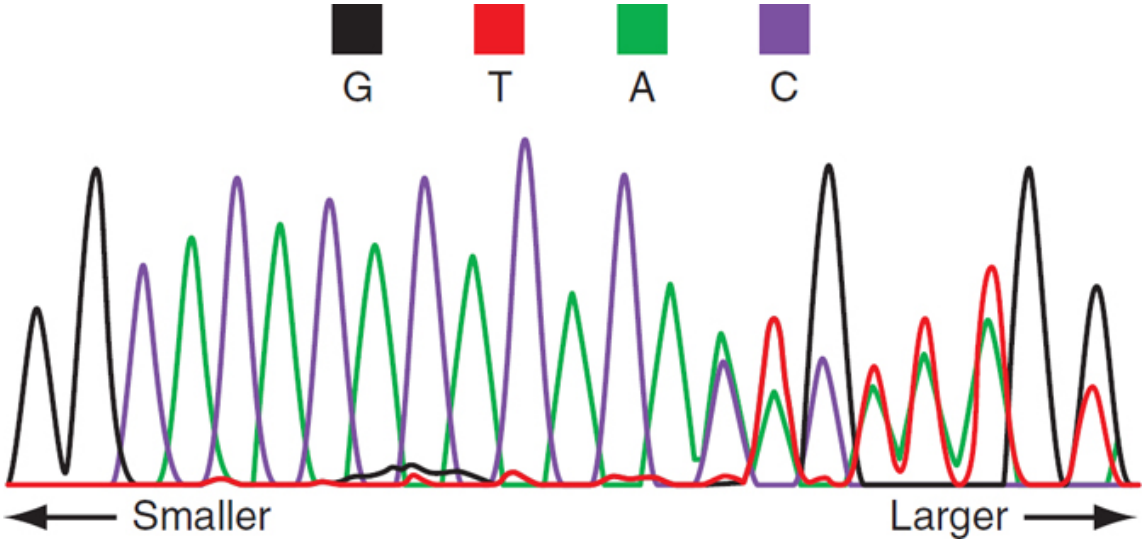
Question

Genomic DNA from a woman's blood cells is PCR amplified by a single pair of primers representing a unique locus in the genome. The PCR products are then sequenced by the Sanger method, using one of the PCR primers as a sequencing primer. The following figure shows a trace of just part of the sequence read.



Question 0.6 pts

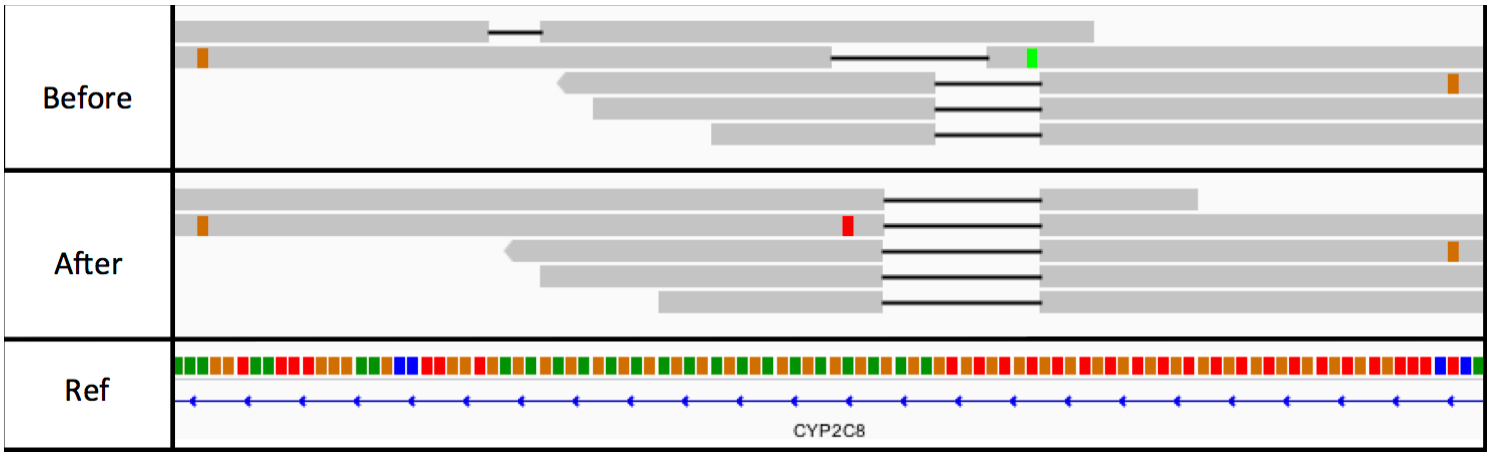
Suppose you wanted to genotype this locus based on single-molecule high-throughput DNA sequencing of whole genomes. A single read aligning to this region in the genome would suffice for genotyping the locus by this alternative method.

True

False

Question 0.6 pts

Which step in the GATK pipeline was run to change the realignments in "Before" to those shown "After"?



Mark Duplicates

Indel Realignment

Question

There are new methods to obtain the whole-genome sequence of a fetus without any invasive procedure such as amniocentesis that could on rare occasions cause miscarriage. This technique is based on the fact that some fetal cells leak into the mother's bloodstream and then break down, releasing their DNA. Assume exactly 10% of the DNA fragments in the mother's blood serum come from the fetus, while the remaining 90% of the DNA fragments in the serum come from the mother's genome.

Investigators collected cell-free DNA from a pregnant woman's bloodstream and subjected it to high-throughput sequencing. The table below looks at seven unlinked loci; the number of reads of particular alleles (identified by Greek letters) are shown. You should assume for the sake of simplicity that all numerical differences are statistically significant (even though actual data are never this clean).

Locus	Sequences	Number of reads
1	α: TCTTTGGTAAACGCAAG	1000
2	α: GTACCGGAGGCAGCCTC	500
	β: GTACCGGCGGCAGCCTC	500
3	α: AGCCATTGCGGATCCGA	950
	β: AGCTATTGCGGATCCGA	50
4	α: GGGGCCTTATGATAAGG	50
5	α: CAGTTCCTGGAGTTGTA	550
	β: CAGTTCATGGAGTTGTA	450
6	α: GCAGCCCGTGCTGTTAA	500
	β: GCAGCCCGTGCTGTCAA	450
7	α: CACTCAGTCCTACGGAC	500
	β: CACTCGGTCCTACGGAC	450
	γ: CACTCAGTCCTAAGGAC	50

Question 0.6 pts

Is the fetus male or female?

Male

Female

Question 0.6 pts

Which locus is X-linked?

1
2
3
4
5
6
7

Question 0.6 pts

What is the genotype of the mother for locus 3?

α/α
α/β
β/β
No answer text provided.

Question 0.6 pts

What is the genotype of the fetus for locus 3?

α/α
α/β
β/β
No answer text provided.

Question 0.6 pts

What is the genotype of the mother for locus 5?

α/α
α/β
β/β
No answer text provided.

Question 0.6 pts

What is the genotype of the fetus for locus 5?

α/α
α/β
β/β
No answer text provided.

Question 0.6 pts

What is the genotype of the mother for locus 7?

α/α
α/β

α/γ
β/γ

Question 0.6 pts

What is the genotype of the fetus for locus 7?

α/α
α/β
α/γ
β/γ

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